

How a lung cell processes and displays an MHC-I peptide

Representative Human context • six biological layers • canonical genes shown for orientation

- 1

Source abundance & lung cell state

EPCAM SFTPA1 SFTPB SCGB1A1
- 2

Proteasomal cleavage

PSMB8 PSMB9 PSMB10
- 3

Transport into the ER

TAP1 TAP2
- 4

ER peptide trimming

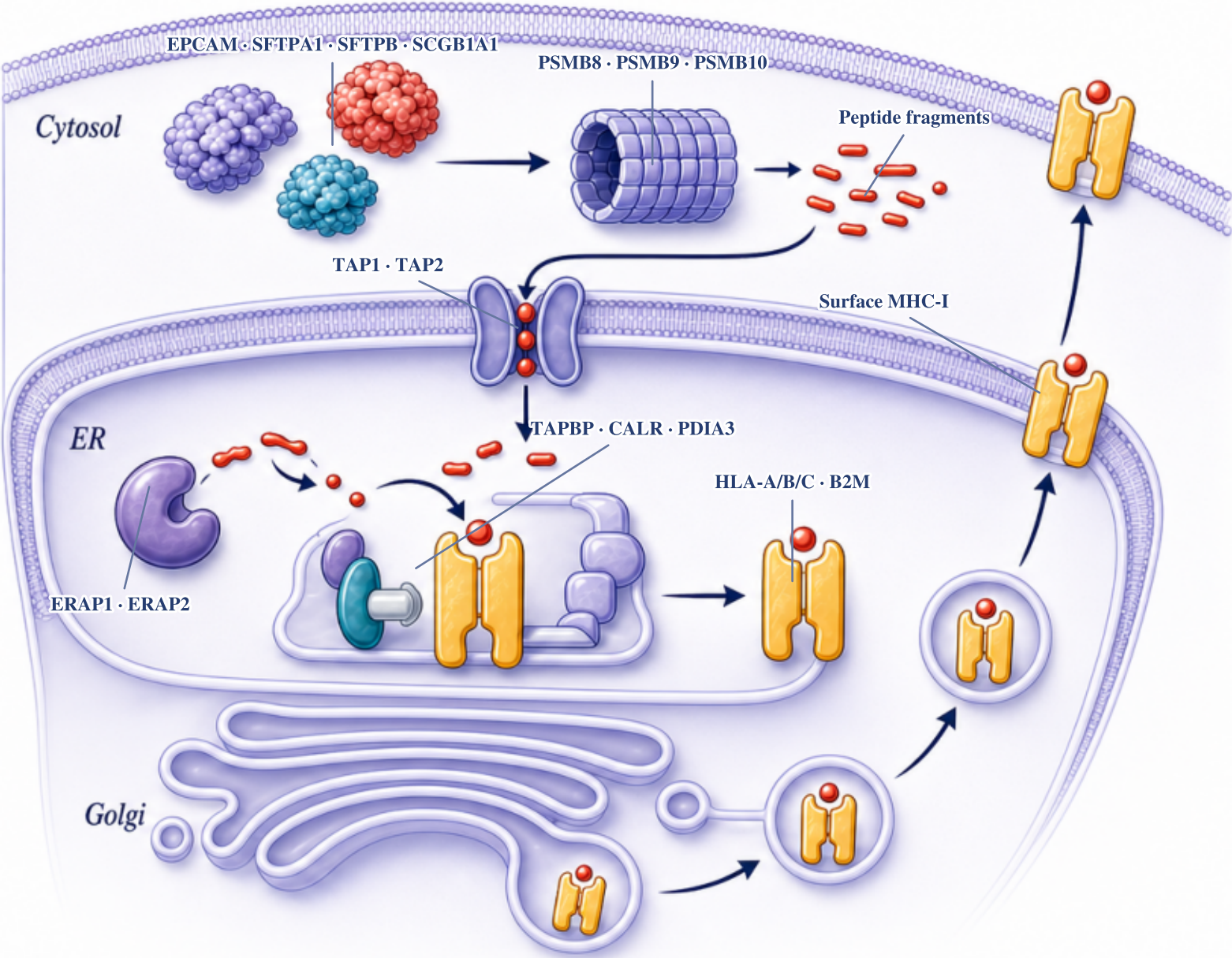
ERAP1 ERAP2
- 5

Peptide-loading complex

TAPBP CALR PDIA3
- 6

Surface MHC-I display

HLA-A/B/C B2M



SOURCE → CLEAVAGE → ER TRANSPORT → TRIMMING → LOADING → SURFACE DISPLAY